



CSE3433 SOFTWARE ARCHITECTURE

PROGRAMME: COMPUTER SCIENCE (SOFTWARE ENGINEERING)

SEMESTER II 2025/2026

GROUP 6

PROJECT TITLE:

SMARTLEARN: CLOUD-NATIVE LEARNING MANAGEMENT SYSTEM (LMS)

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Weekly Progress

PHASE 1 (Week 3 – Week 5) – Project Proposal

Week	Activities Undertaken	Individual Contribution	Challenges	Action Plan
Week 3	Initial system idea & concept formation	Contributed frontend perspective to the SMARTLEARN concept and discussed client-tier layout requirements.	Unclear system scope and system direction.	Research real-world cloud-native LMS structures. Break down the system into user management, course management, and system flow modules. Use AI tools to validate concept clarity and ensure project feasibility before designing.
Week 4	Requirement analysis & project validation	Researched user journey paths and suggested essential student/lecturer UI view models	Difficulty narrowing system scope.	Analyze similar LMS platforms and extract only essential features for academic implementation. Separate functional and non-functional requirements clearly. Validate the system direction with teammates and refine the scope with the lecturer to avoid over-complexity.
Week 5	Proposal drafting & finalisation	Outlined frontend-specific functional needs like login gates, course card views, and enrollment modules.	Time pressure before submission deadline.	Review the entire proposal structure (objectives, scope, requirements, and methodology) for consistency. Proofread, correct formatting errors, and perform a final validation before submission to ensure completeness and compliance.

PHASE 2 (Week 6 – Week 7) – Architecture Design

Week	Activities Undertaken	Individual Contribution	Challenges	Action Plan
Week 6	Researching split integration models between client views and server APIs.	Designed secure fallback route patterns to block users from jumping straight to dashboards without logging in.	Lack of familiarity with handling distributed asynchronous data streams.	Learn how to cleanly decouple static HTML assets away from runtime Node.js server dependencies.
Week 7	Refining user system workflow interactive architectural diagrams.	Mapped step-by-step user click pathways tracing from the initial login form to the grade list views.	Main workflow arrows becoming overly complex and confusing to follow.	Streamline chart paths by focusing strictly on the core operations of the primary system roles.

PHASE 3 (Week 8 – Week 11) – Prototype Progress

Week	Activities Undertaken	Individual Contribution	Challenges	Action Plan
Week 8	Setting up local development environment to launch the prototype code base.	Built custom CSS responsive grids to anchor buttons and form text alignments securely.	Layout grids broke and overlapped on small viewports; accidentally toggled off the main file explorer.	Prompt Gemini for terminal navigation commands, activate auto-save preference keys, and utilize Flexbox properties to fix mobile views.
Week 9	Coding Authentication (Auth) properties and client-side scripting logic	Programmed async/await fetch blocks to dispatch user credentials and handle JWT storage.	Verification routines failed to authorize payloads; token values dropped completely upon page refresh loops.	Establish an active client authentication middleware routine checking active token keys within local storage pools.
Week 10	Prepping production builds for deployment onto cloud infrastructures (Render & Vercel).	Cleaned up local repository commits and investigated environment hosting dashboard controls.	Remote deployment pipelines crashed; encountered severe CORS blocks trying to cross-communi	Ask Gemini for guidance on setting Environment Variables (ENV) inside Render and structuring vercel.json routing rewrites.

			cate distributed nodes.	
Week 11	Debugging UI components across responsive mobile view screens.	Implemented custom horizontal masking wrappers onto academic grade listings to avoid display text clipping.	Encountered a continuous 'Failed to fetch' network rule error on the production node	Locate and replace hardcoded localhost backend URL strings with correct production Render api server links.

PHASE 4 (Week 12 – Week 13) - Project Report, Individual Logbook & Final Demo

Week	Activities Undertaken	Individual Contribution	Challenges	Action Plan
Week 12	Conducting end-to-end full network testing across live deployed cloud infrastructure instances.	Loaded mock payload structures into data forms to verify field durability against remote database engines.	Debugging errors became extremely slow due to massive, convoluted terminal stack traces.	Utilize Gemini to isolate and clarify error roots; adjust client request header formats to correctly match JSON formats.

Week 13	Packaging engineering logs and final presentation simulation flows.	Reformatted historical AI tracking records for the logbook sheets and polished UI showcase scripts.	Logbook logs were overly dense, cluttered, and contaminated with external teammate sample templates.	Purge external template entries, isolate personal development prompts, and refine descriptions down into clear summaries.
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GenAI Usage Record

Week	Date	Tool Used	Purpose	Prompt Used	Output Summary	How It Was Used
Week 8	11/05/2026	Gemini	Editor Environment Configuration	"Weh, aku first time guna Visual Code ni, nak kena tekan tang mana eh bagi coding ni jalan?"	Guided on basic navigation panel functions and main workspace windows.	Used to familiarize with VS Code workspace boundaries.
Week 8	12/05/2026	Gemini	Editor Terminal Management	"Butang terminal kat Visual Code ni kat mana? Macam mana nak run codingNode.js kat situ?"	Indicated the shortcut combination keys and input executable syntax loops.	Used to open execution layer window and trigger server test nodes.

Week 9	18/05/2026	Gemini	Engine Setup & Scaffolding	"Macam mana nak tulis coding untuk Node.js bagi sistem backend? Aku first time guna ni."	Supplied boilerplate files detailing index script boundaries and route hooks.	Used to understand structural baseline elements for runtime processing.
Week 9	19/05/2026	Gemini	Environment Initialization	"Macam mana nak install package npm? Kena taip kat terminal VS Code ka?"	Outlined the package execution commands utilizing standard runtime setups.	Applied inside local terminals to cleanly assemble essential dependencies.
Week 9	20/05/2026	Gemini	Component Communication	"Macam mana nak link file JavaScript dengan HTML? Aku dah letak tapi asal butang login tak ada respon bila tekan?"	Indicated missing client script loading order and binding elements.	Repositioned tag logic properties to capture login interactive triggers.

Week 9	22/05/2026	Gemini	Component Security Construction	"Macam mana nak buat coding auth guna JWT token untuk sistem login?"	Modeled encryption layer signatures for verifying incoming users.	Used to safely establish foundational identification checking functions.
Week 9	23/05/2026	Gemini	Route Protection Framework	"Aku nak buat middleware dalam Node.js untuk check authentication user, macam mana coding dia?"	Supplied interceptor structures inspecting incoming headers for matching credentials.	Implemented on secure routes to automatically block anonymous entities.
Week 10	29/05/2026	Gemini	Source Version Management	"Weh help, folder node_modules ni kena upload sekali kat GitHub ka?"	Indicated standard .gitignore rules mapping out explicit artifact boundaries.	Generated local tracking exclusion sheets to safely shield code repositories.
Week 10	30/05/2026	Gemini	Cloud Service Provisioning	"Macam mana nak deploy projek dalam Render? Kat dashboard tu butang mana yang kena tekan?"	Provided clear UI maps to start live services via active control nodes.	Followed control patterns to connect live remote hosting branches.

Week 11	01/06/2026	Gemini	Cloud Client Provisioning	"Kat dashboard Vercel ni, butang mana nak kena tekan untuk start import projek dari GitHub?"	Illustrated structural pipeline setups to dynamically ingest remote tracking links.	Used to trigger front-end generation patterns on production nodes.
Week 11	02/06/2026	Gemini	Component Connectivity Design	"Weh, macam mana nak connect frontend kat Vercel dengan backend kat Render? Aku try panggil API tapi tak lekat."	Indicated endpoints updates tracing directly onto server domains.	Updated API configuration arrays to trace live distributed components.
Week 11	03/06/2026	Gemini	Cross-Domain Network Diagnostics	"Kenapa bila aku frontend aku kat Vercel panggil API backend dekat Render, dia terus keluar CORS error? Macam mana nak set?"	Detailed origin validation mismatches and suggested header permissions.	Handed configuration blocks over to backend leads to relax server rules.

Week 11	04/06/2026	Gemini	Transport Layer Debugging	"Kenapa keluar error 'Failed to fetch' kat inspect element browser ni? Penat dah aku adjust."	Diagnosed active endpoint mismatch calls executing local network ranges.	Replaced lingering local parameters with correct target resource links.
Week 11	05/06/2026	Gemini	Production Route Error Fix	"Weh, kenapa kat localhost elok ja boleh tekan, tapi bila dah deploy kat Vercel dia keluar page '404 Not Found'?"	Outlined client-side route fallback definitions via rewritten resource parameters.	Implemented custom routing definitions inside Vercel setup configuration sheets.
Week 11	06/06/2026	Gemini	Database Link Verification	"Macam mana nak check coding backend aku ni dah connect dengan database ke belum? Ada keluar error [tepek error]."	Isolated missing environment parameters blocking operational connection tasks.	Configured server boot blocks to cleanly capture tracking messages.

Week 12	08/06/2026	Gemini	Stack Validation Tasks	"Kenapa bila aku tambah data baru kat frontend, dia tak masuk pun dalam database? Tang mana yang tak bersambung ni?"	Indicated payloads translation format errors and missing validation parsers.	Fixed client fetch header definitions to accurately map payload attributes.
Week 12	09/06/2026	Gemini	Execution Diagnostics	"Kenapa keluar error macam ni? [Tepek error] Pening lah macam mana nak fix?"	Decoded runtime memory crashes triggered by unhandled method properties.	Corrected internal parameter paths to guarantee system structural continuity.
Week 13	15/06/2026	Gemini	Information Compression Task	"PLEASE SIMPLIFIED IT , THIS IS TOO MUCH INFORMATION"	Streamlined long operational tracking descriptions into compact scannable elements.	Applied to condense multi-page technical journals down into standard sizes.